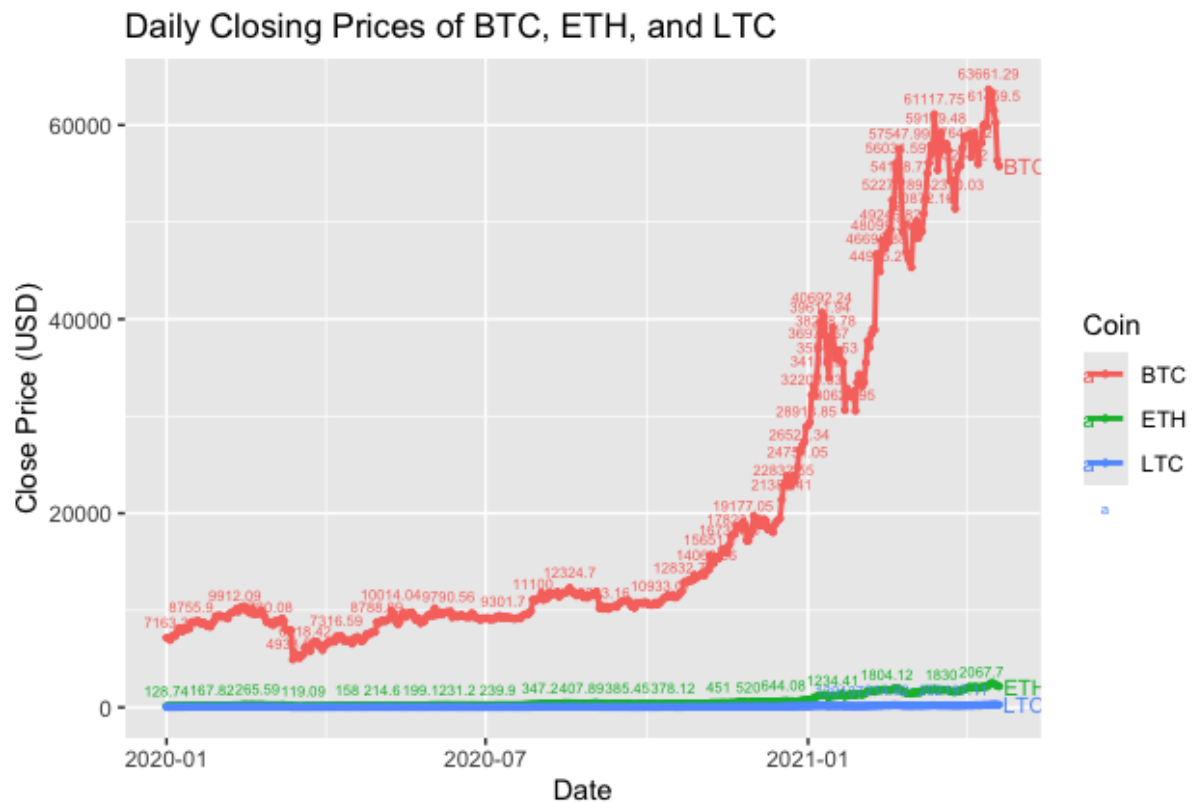


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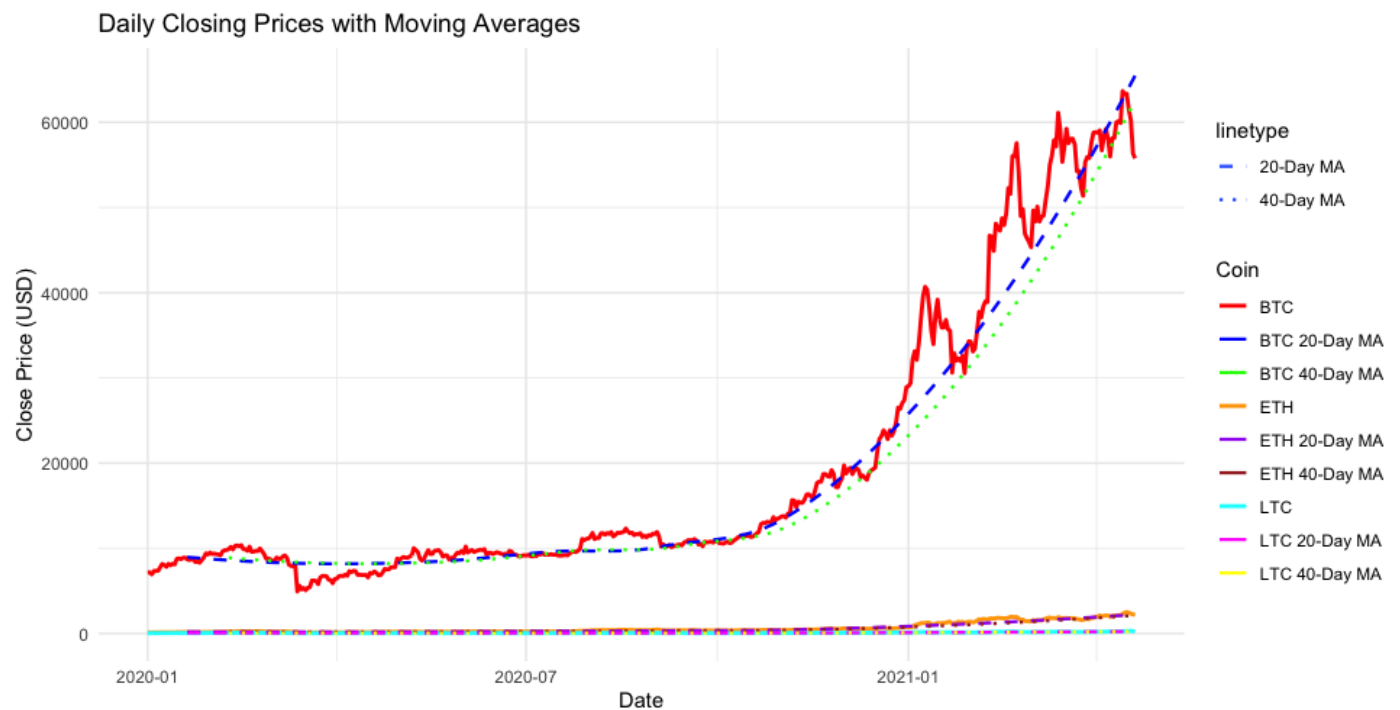
Part A:

This chart visualizes the daily closing prices of Bitcoin (BTC), Ethereum (ETH), and Litecoin (LTC) from January 2020 to April 2021. Numeric labels display the exact closing prices for each day, while line labels identify the respective cryptocurrencies, illustrating the price trends and variations over the period.



Part B:

This chart includes the daily closing prices of Bitcoin (BTC), Ethereum (ETH), and Litecoin (LTC) along with their 20-day and 40-day moving averages. LOESS smoothing is applied to the moving averages, and distinct colors differentiate the moving averages, highlighting short- and long-term trends in the price data.



The whole code:

```
library(ggplot2)
library(readr)
library(dplyr)

btc <- read_csv("~/Desktop/btc_daily.csv")
eth <- read_csv("~/Desktop/eth_daily.csv")
ltc <- read_csv("~/Desktop/ltc_daily.csv")

btc <- btc %>% mutate(Coin = "BTC")
eth <- eth %>% mutate(Coin = "ETH")
ltc <- ltc %>% mutate(Coin = "LTC")

# Bind all data into one dataframe
all_data <- bind_rows(btc, eth, ltc)
all_data$Day <- as.Date(all_data$Day)

all_data <- all_data %>%
  group_by(Coin) %>%
  mutate(Normalized_Close = (Close - min(Close)) / (max(Close) - min(Close)))

# Part A
ggplot(all_data, aes(x = Day, y = Close, color = Coin)) +
  geom_line(size = 1) +
  geom_text(data = filter(all_data, Day == max(Day)),
    aes(label = Coin), hjust = -0.1, size = 3) +
  geom_point(size = 0.8) +
  geom_text(aes(label = round(Close, 2)), vjust = -1, size = 2, check_overlap = TRUE)
labs(title = "Daily Closing Prices of BTC, ETH, and LTC",
  x = "Date",
  y = "Close Price (USD)")
theme_minimal() +
  theme(legend.position = "none")

# Part B
ggplot(all_data, aes(x = Day, y = Close, color = Coin)) +
  geom_line(size = 1) +
  geom_smooth(aes(y = MA_20, linetype = "20-Day MA", color = paste(Coin, "20-Day MA")),
    method = "loess", se = FALSE, size = 0.8) + # 20-day moving average
  geom_smooth(aes(y = MA_40, linetype = "40-Day MA", color = paste(Coin, "40-Day MA")),
    method = "loess", se = FALSE, size = 0.8) + # 40-day moving average
  scale_color_manual(values = c(
```

```
"red", "blue", "green",      # BTC, ETH, LTC lines
"orange", "purple", "brown", # BTC, ETH, LTC 20-Day MA
"cyan", "magenta", "yellow"  # BTC, ETH, LTC 40-Day MA
)) +
scale_linetype_manual(values = c("20-Day MA" = "dashed", "40-Day MA" = "dotted")) +
labs(title = "Daily Closing Prices with Moving Averages",
      x = "Date",
      y = "Close Price (USD)") +
theme_minimal()
```